Does Removing Non-Anatomical Artifacts from Chest X-Rays Improve Pneumonia Classification?

AAI Exam Project
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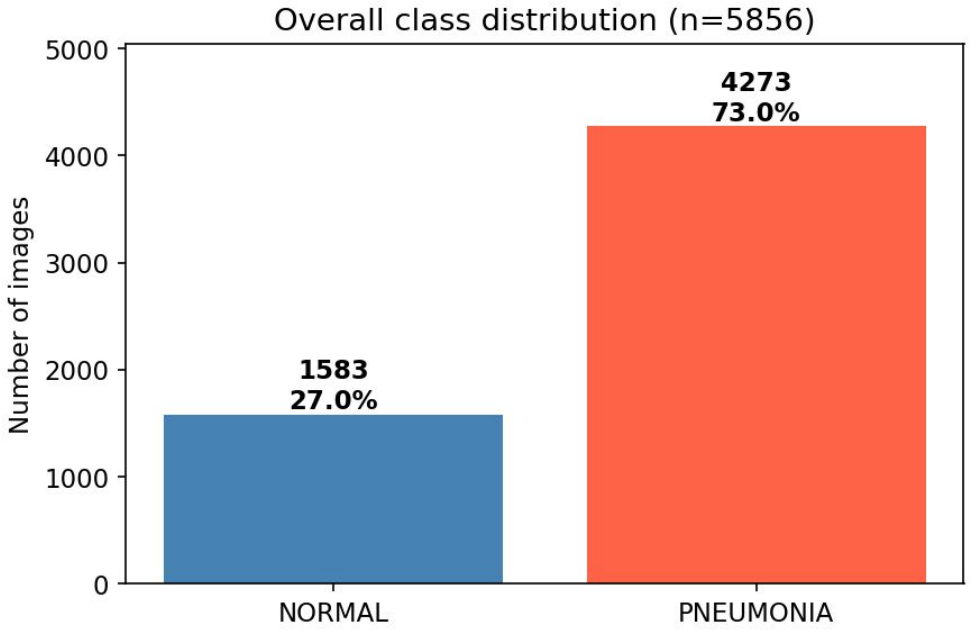
Motivation

Shortcut Learning:

Chest X-rays carry burned-in orientation markers (R/L) and other non-anatomical artifacts that CNN might exploit as a shortcut instead of learning of genuine pathology.

We ask whether removing these artifacts improves pneumonia classification.

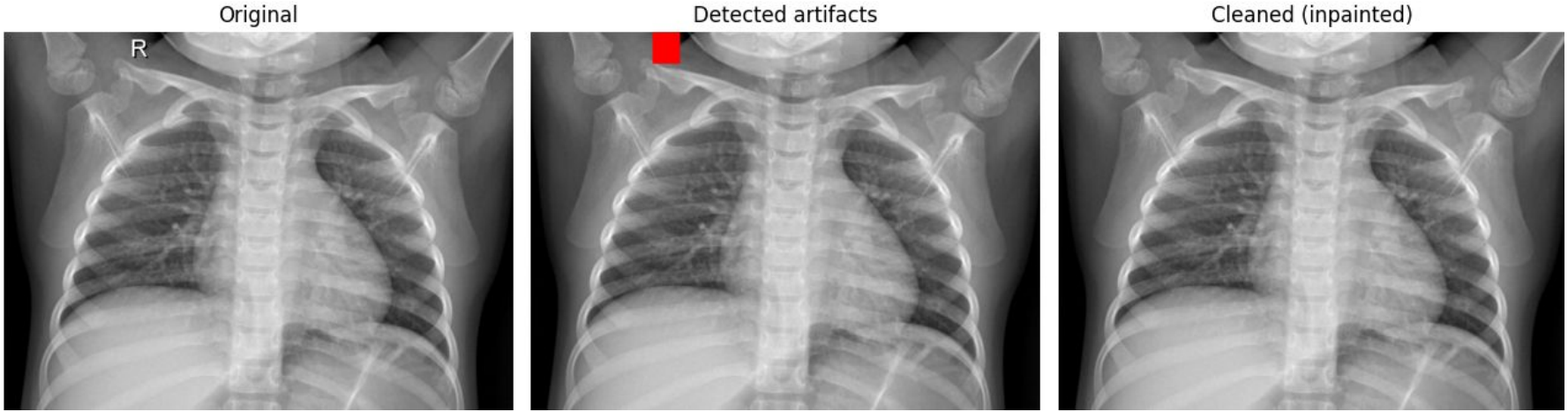
Data



- Kaggle *Chest X-Ray (Pneumonia)* dataset
- 5856 chest x-ray images
- 73/27 class imbalance (more pneumonia)*
- Original had only 16 img val set ➡ Re-split 80/10/10 train/val/test (same split for both experiments)
- We create another dataset of cleaned images, that is used for the experiment.

*WeightedRandomSampler balances training batches (val and test keep their natural prevalence)

Artifact Removal



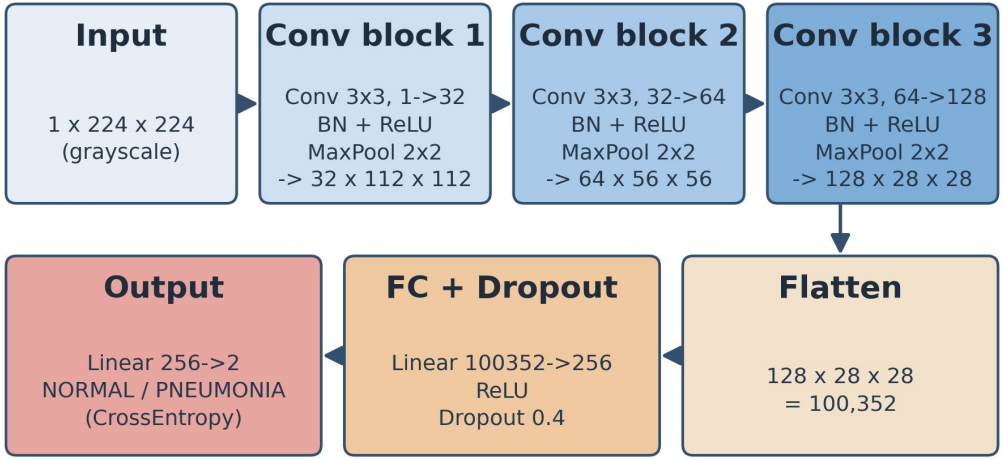
- OpenCV 3-test heuristic (Location, shape, isolation)
- ~85% of images had something detected and inpainted
- Telea inpaint (no hard edge like black box, no loss of anatomy like crop)

Pipeline

Same pipeline runs on both datasets, preprocessing is identical.

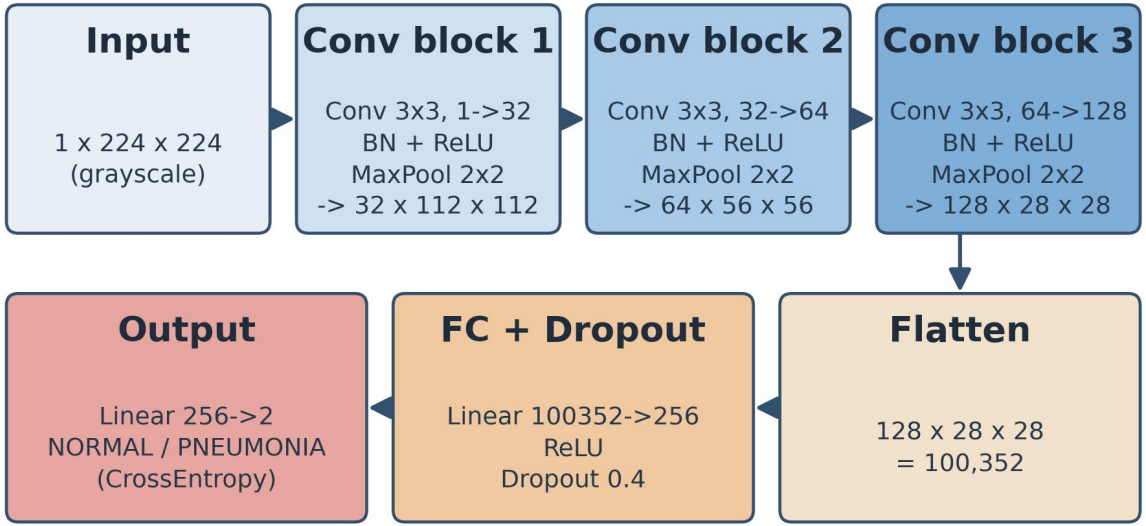
- Greyscale (1 channel)
- Pad square to avoid stretch
- Resize to 224x224
- Light augmentation of train set only (Rotation ±7°, shift ±3%, zoom ±3%, no h-flip)
- Normalise[0,1] to [-1, 1]
- Fixed seed random generators

Model



- Custom three-block CNN.
- Input is 32x1x224x224 tensor (batch size of 32 x 1 channel x ~50k values per image).
- Flatten reshapes the 3D feature map into a 1D vector.
- Fully Connected NN (~100k inputs, 1 hidden layer+ReLU)
- Dropout 40% in training.
- Model outputs two logits (SoftMax applied later)

Model: Convolution Blocks



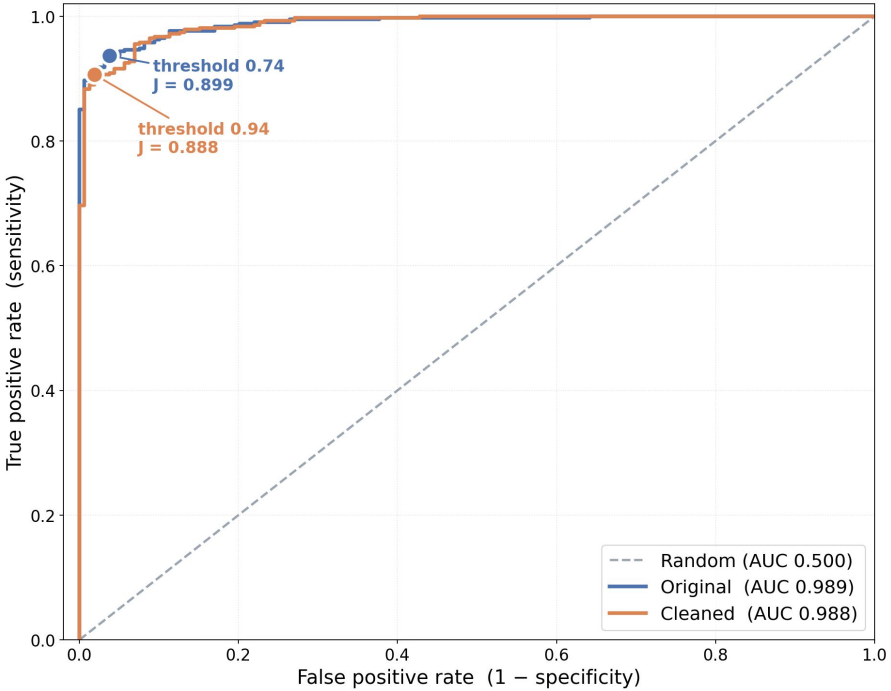
Convolution finds a pattern anywhere in the image, pooling makes the answer less sensitive to position.

- Each conv layer slides a small 3x3 filter across its input producing one feature map per filter.
- Early layers detect edges, gradients and texture whereas later layers learn larger structures.
- Each subsequent block **doubles the filter count** and **halves the resolution**.
- Each layer applies normalization to keep activations well scaled.

Experiment Design

- Two fresh models are trained from scratch, one per dataset, both with the same seed.
- Locked across both runs:
  - **Data:** Split membership, preprocessing, sampler logic
  - **Training:** Architecture, AdamW, ReduceLROnPlateau, 15 epochs (best epoch by validation Youden's J), CrossEntropyLoss and hyperparameters
  - **Evaluation:** Threshold selection on validation by Youden's J
- *Only the dataset path differs.*
- Hyperparameter tuning: Tuned config won on val but lost on test to default, so default was used (test J 0.8835 versus 0.8868).
  - Default config: lr 0.0001, weight decay 0, dropout 0.4, batch size 32
- Experiment is reproducible to ~3 decimals on GPU, not bit identical.

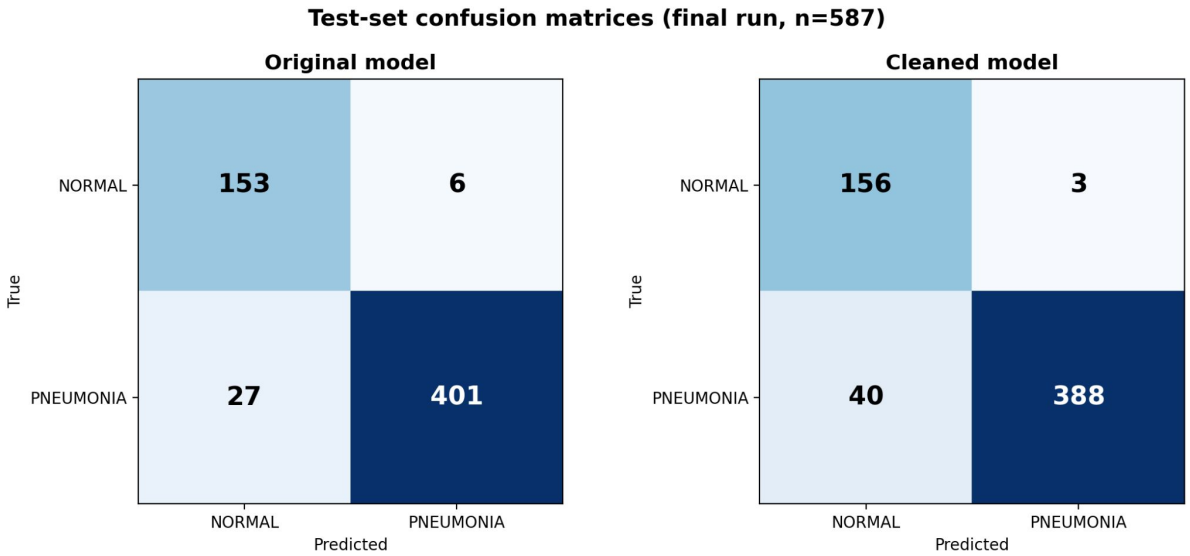
Performance Metrics



- Accuracy is wildly misleading with datasets that have large class imbalance.
- **ROC/AUC** is immune to this because each axis is normalized within its own true class (patient's real condition).
- **Youden's J** (sensitivity + specificity -1) is the vertical distance from a point on the curve down to the diagonal (0 is random, 1 is perfect).
- Threshold on val, applied once

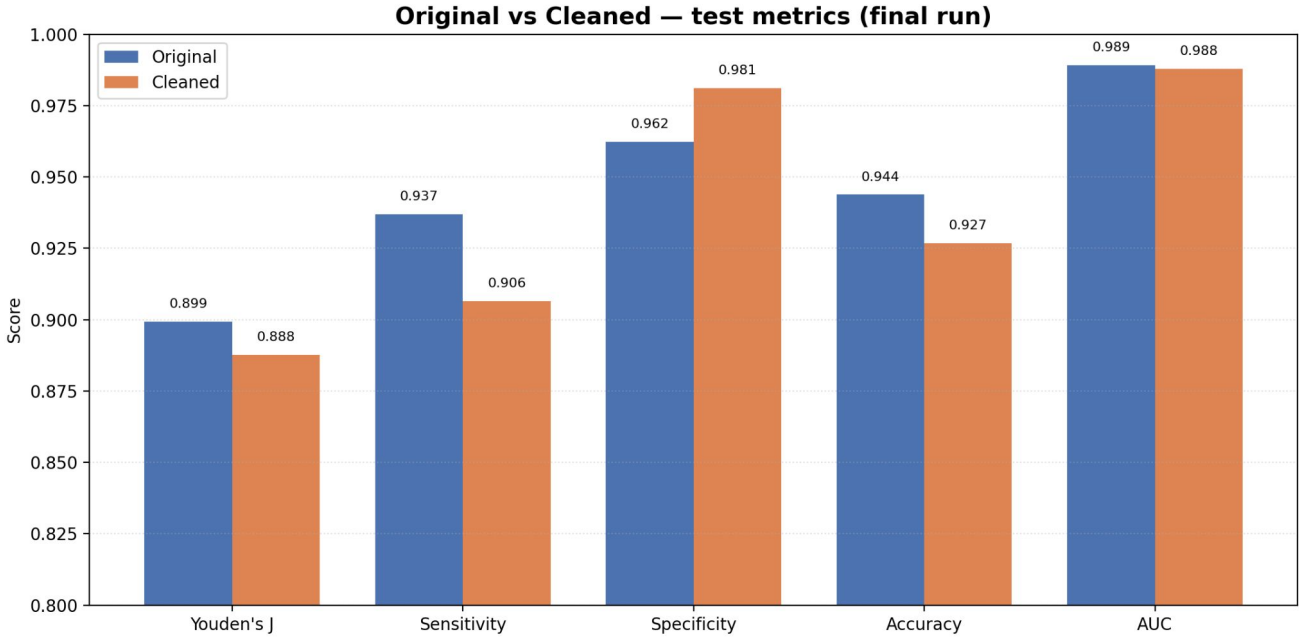
AUC judges the model, Youden's J chooses the prediction threshold.

Results: Confusion Matrix



- Test set n = 587 (159 normal, 428 pneumonia), only touched once.
- Cleaned model's more conservative decision threshold shows up as some more false negatives (missed pneumonia cases) and a few less false positives.

Results: All Metrics



Sensitivity=TP rate, Specificity=TN rate

- *Model's ability to discriminate is unchanged.* AUC 0.989 → 0.988 (−0.0013).
- Youden's J change (-0.0115) reflects the change in sensitivity and specificity.
- Sensitivity −3.0pp, specificity +1.9pp (tradeoff from higher threshold).
- Cleaning did not improve classification.

Interpretation

- Removing non-anatomical artifacts did not improve pneumonia classification.
- No evidence that the model relied on non-anatomical markers.
- Decisions plausibly already driven by lung fields (markers sit outside of lung fields, and sampler + augmentation already dilute them).
- The cleaned dataset is not 100% clean, which might dilute the result.
- The markers in this single-source dataset might not correlate with the class.
- Behavioural evidence only (see future work).

Future Work

- Grad-CAM saliency maps for visual evidence
- k-fold confidence intervals for better estimate
  - Patient grouped would keep all images from same patient in one fold (prevents leakage)
- U-Net lung segmentation
- Quantify detector misses
- Global average pooling instead of flatten

The End